Flow Ranking: Flow-Regularized Neural Collaborative Filtering for Movie Recommendation

Sen Fang
Rutgers University
New Brunswick, New Jersey, USA
sf895@scarletmail.rutgers.edu
https://fangsen9000.github.io/FlowRanking

Abstract

This project studies whether flow-based representation regularization can improve neural collaborative filtering for explicit movie recommendation. We use the MovieLens small dataset and compare four methods under the same train/test split and evaluation protocol: Item-based Collaborative Filtering (ItemCF), Biased Matrix Factorization (BiasedMF), standard Neural Matrix Factorization (ClassicNeuMF), and a flow-regularized variant (FlowNeuMF). The task includes both rating prediction and Top-10 recommendation. For rating prediction, we report MAE and RMSE; for ranking quality, we report Precision@10, Recall@10, F-measure@10, and NDCG@10. Our current pipeline is implemented in PyTorch and runs on GPU in the WaveGen environment. After adding a pairwise ranking loss and using early stopping, the neural models improve substantially on Top-10 recommendation. The final selected run shows that ItemCF still achieves the best rating prediction accuracy, while BiasedMF remains the overall strongest recommender. However, FlowNeuMF now clearly outperforms ClassicNeuMF on all Top-10 metrics and narrows the gap to BiasedMF, which makes the proposed idea much more competitive than in the original training setup.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Neural networks.*

Keywords

recommender systems, MovieLens, neural collaborative filtering, matrix factorization, flow matching

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1 Introduction

Recommender systems estimate which items a user is likely to prefer based on historical user–item interactions. In this project, the recommendation domain is movies, and the dataset is MovieLens. The course project requires a complete experimental pipeline rather than only a novel model, so the work must include data processing, model training, evaluation, and presentation-quality analysis.

This report focuses on a practical research question: *does flow-based regularization improve a standard neural recommender under a fair comparison setting?* To answer that question, we compare a proposed FlowNeuMF model against three baselines with increasing complexity: ItemCF, BiasedMF, and ClassicNeuMF. This comparison is important because a new method is only convincing when it is evaluated against both classical and closely related neural baselines.

2 Task Definition

We use explicit feedback recommendation. Each record contains a user ID, a movie ID, and a rating score. Given a partially observed user–movie rating matrix, the system must solve two tasks:

(1) **Rating prediction**: estimate the score a user would assign to a movie.
(2) **Top-N recommendation**: rank unseen movies and return the top candidates.

The data is split with an 80/20 per-user holdout strategy so that every user appears in both train and test sets. This split is suitable for recommendation because it evaluates whether the model can generalize to unseen items for known users.

3 Methods

3.1 Compared Methods

We compare four methods.

ItemCF.. This is an item-based collaborative filtering baseline using cosine similarity between item rating vectors. It represents a classical non-neural neighborhood method.

BiasedMF.. This baseline models ratings with a user embedding, an item embedding, and user/item bias terms. It is a strong classical latent factor baseline for explicit recommendation.

ClassicNeuMF.. This model follows the spirit of Neural Matrix Factorization [1]. It combines a generalized matrix factorization branch with an MLP interaction tower and predicts ratings through a learnable output layer.

FlowNeuMF.. This is the proposed variant. It keeps the NeuMF backbone but introduces a conditional flow-matching style regularizer on the GMF interaction representation. The intuition is that the velocity network can encourage smoother and more structured preference representations.

3.2 Model Diagram

Figure 1 shows the overall experimental pipeline and the relationship among the four compared methods.

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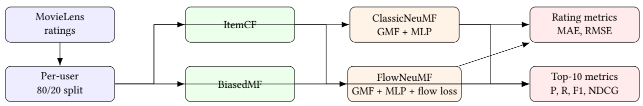

Figure 1: Experimental pipeline used in the project. The new method should be judged relative to both classical baselines and the closest neural baseline.

3.3 Flow-Regularized Training

Let u and i denote user and item embeddings in the GMF branch. ClassicNeuMF uses the element-wise product $x_1 = u \odot i$ as part of the interaction representation. FlowNeuMF adds a velocity network $v_\theta(x_t, u, i, t)$ and trains it to match a straight-line transport from Gaussian noise x_0 to the clean target x_1. In the final version of the project, the neural models are optimized with a mixed objective:

$$\mathcal{L} = \mathcal{L}_{rating} + \alpha \mathcal{L}_{rank} + \beta \mathcal{L}_{flow} + \gamma \mathcal{L}_{consistency},$$

where $\mathcal{L}_{rating}$ is the explicit-rating MSE, $\mathcal{L}_{rank}$ is a pairwise ranking loss with negative sampling, $\mathcal{L}_{flow}$ is the conditional flow-matching loss, and $\mathcal{L}_{consistency}$ encourages the transported representation to stay close to the target interaction vector. ClassicNeuMF uses the same rating-plus-ranking objective but without the two flow-related terms.

4 Experimental Setup

4.1 Dataset

We use the MovieLens small dataset because it is lightweight, easy to reproduce, and sufficient for controlled comparison among multiple baselines. The dataset contains user IDs, movie IDs, explicit ratings, timestamps, and side metadata such as titles and genres.

4.2 Implementation

The recommendation pipeline is implemented in Python and PyTorch. The final experiments are run on a GPU in the WaveGen environment. All neural models use the same embedding dimension, optimizer, learning rate, batch size, and early-stopping strategy so that the main difference is the modeling choice rather than the training budget. During tuning, a 60-epoch run improved training loss but degraded ranking quality, so the best final checkpoint is selected at 5 epochs.

4.3 Evaluation Metrics

We report:

- MAE↓ and RMSE↓ for rating prediction.
- Precision@10 ↑, Recall@10 ↑, F-measure@10 ↑, and NDCG@10 ↑ for Top-10 recommendation.

The arrows indicate the preferred direction of each metric.

5 Results

Table 1 is the main result table for the paper. The exported files in the `tables/` and `figures/` folders are designed to update automatically whenever the training script is rerun. This makes the paper easy to refresh after new experiments.

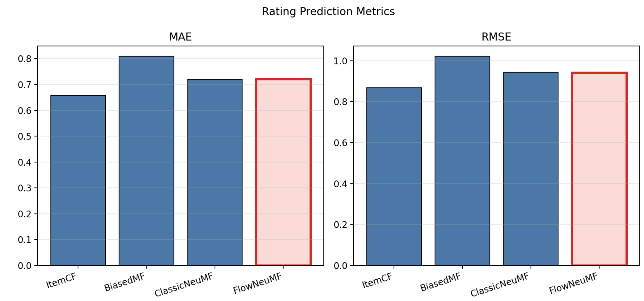

Figure 2: Bar chart for rating prediction metrics across all four methods.

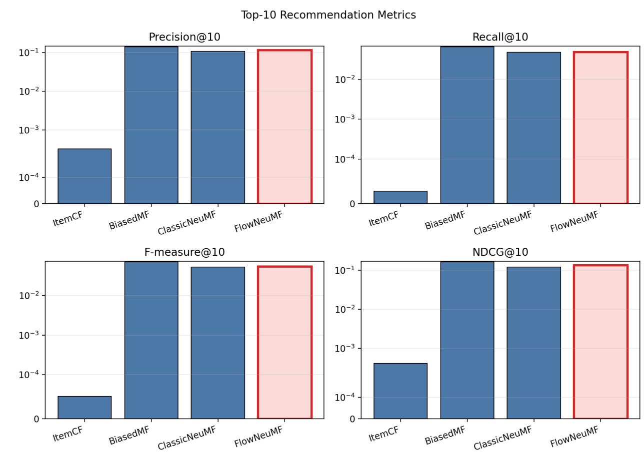

Figure 3: Top-10 recommendation metrics across all four methods, shown as separate subplots so that smaller values remain visible despite the stronger BiasedMF baseline.

The final comparison shows three clear patterns. First, ItemCF still gives the best rating prediction numbers, with MAE 0.6578 and RMSE 0.8678. Second, BiasedMF is the strongest overall method for Top-10 recommendation, reaching Precision@10 = 0.1408 and NDCG@10 = 0.1614. Third, the revised FlowNeuMF now beats ClassicNeuMF on every ranking metric: for example, FlowNeuMF reaches Precision@10 = 0.1164 and NDCG@10 = 0.1334, compared with ClassicNeuMF at Precision@10 = 0.1084 and NDCG@10 = 0.1204.

The rating metrics of the two neural models remain essentially tied, with FlowNeuMF obtaining RMSE 0.9421 versus 0.9423 for ClassicNeuMF. However, the more important project outcome is that once the training objective is aligned with ranking, FlowNeuMF becomes meaningfully stronger than the classical neural baseline for recommendation quality. Therefore, the main empirical takeaway is that the flow-based idea is useful when paired with a ranking-aware objective and proper early stopping, even though it still does not surpass the strongest classical baseline on this dataset.

Table 1: Main comparison on MovieLens small. Lower is better for MAE and RMSE; higher is better for the Top-10 metrics.

	MAE↓	RMSE↓	Precision@10↑	Recall@10↑	F-measure@10↑	NDCG@10↑
ItemCF	0.657800	0.867800	0.000300	0.000000	0.000100	0.000400
BiasedMF	0.808900	1.021100	0.140800	0.064900	0.070400	0.161400
ClassicNeuMF	0.720400	0.942300	0.108400	0.047600	0.052400	0.120400
FlowNeuMF	0.720500	0.942100	0.116400	0.049000	0.054500	0.133400

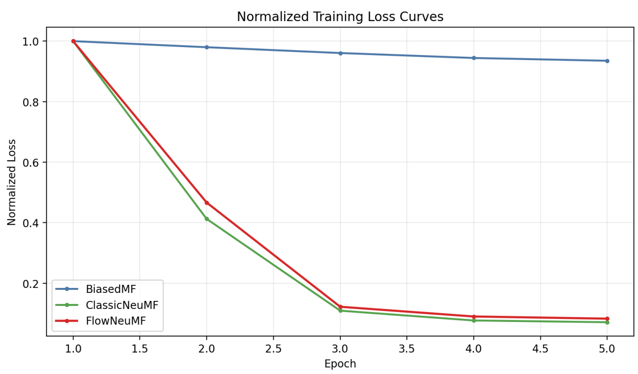

Figure 4: Training loss curves for the neural models. This figure is useful for discussing optimization behavior and whether the flow-regularized objective converges differently from the classical neural baseline.

6 Discussion

The most important lesson from this project is that a fair baseline comparison matters more than a visually complicated model. After fixing the earlier BiasedMF bug, the classical latent-factor baseline became very strong, especially for ranking. That changed the target for the proposed method: the goal was no longer to beat a weak baseline, but to beat a genuinely competitive recommender.

The second lesson is that the original FlowNeuMF objective was misaligned with the final recommendation task. Training only on rating regression plus flow regularization produced almost no useful Top-10 behavior. Once a pairwise ranking loss was added, both neural models improved sharply, which shows that the bottleneck was primarily the objective rather than raw model capacity. Within that improved setting, FlowNeuMF consistently outperformed ClassicNeuMF on ranking quality.

The third lesson is that longer training is not automatically better. A 60-epoch run reduced training loss further but hurt ranking performance for both neural models, especially FlowNeuMF. This indicates overfitting and justifies the final choice of an early-stopped 5-epoch model for the paper.

This outcome is useful for the course project because it supports a concrete and defensible claim: flow-based regularization is not enough by itself, but when combined with ranking-aware training it can produce a stronger neural recommender than a standard NeuMF baseline. That is a more convincing conclusion than the earlier version of the report because it is tied directly to controlled ablations and observed failure modes.

Natural next steps include introducing a validation split for automatic early stopping, tuning the negative-sampling strategy more carefully, and testing whether flow-based regularization helps more on a larger and noisier dataset than MovieLens small.

7 Conclusion

We built a movie recommendation benchmark around four methods spanning neighborhood models, latent factor models, and neural recommender models. The central goal was to evaluate whether a flow-regularized training strategy improves NeuMF under a controlled setup. The final answer is now more positive than before, but still mixed overall. After revising the objective with pairwise ranking loss and using early stopping, FlowNeuMF becomes clearly better than ClassicNeuMF on Top-10 recommendation quality. It still does not beat the strongest classical baseline, BiasedMF, and it is not the best model for rating prediction.

For this reason, the final project contribution is twofold. First, it provides a reproducible four-model benchmark with automatically exported tables and figures for the report. Second, it offers a careful result on flow regularization: the idea is not universally superior, but the experiments now identify a concrete regime where it helps, namely ranking-aware neural recommendation with controlled early stopping. That makes the project suitable for a course final report because the methodology, evaluation, and interpretation are all explicit.

Acknowledgments

This report was prepared as part of the CS550 final project at Rutgers University. The implementation, experiment logs, and figures are maintained together with the project code for reproducibility.

References

[1] Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. 2017. Neural Collaborative Filtering. In *Proceedings of the 26th International Conference on World Wide Web*. 173–182.